XRAY FUNDAMENTALS AND HANDS-ON PRACTICE – STUDENT MANAGEMENT SYSTEM PROJECT

1. Introduction to Jira and Xray

1.1 What is Jira?

Jira is a project management tool used by software development teams to plan, track, and manage their work.

It helps teams organize tasks, track project progress, manage requirements, and identify defects.

1.2 What is Xray?

Xray is a test management application integrated with Jira. It helps testing teams manage test cases, test plans, test sets, test executions, and test results.

1.3 Why Xray is Used

Xray helps teams to:

- Create and manage test cases.
- Organize test cases using Test Sets.
- Prepare Test Plans.
- Execute test cases and record results.
- Manage preconditions.
- Track testing progress.
- Generate test reports.

2. Understanding Software Testing

2.1 What is Software Testing?

Software testing is the process of checking whether an application works according to its requirements.

It helps identify defects and ensures that the application behaves as expected.

2.2 Types of Testing

Important types of testing include:

- Functional Testing
- Manual Testing
- Regression Testing
- Integration Testing
- System Testing
- Acceptance Testing

2.3 Manual Testing

Manual testing involves executing test cases manually without using automation tools.

In this project, manual testing was performed for the Student Login functionality.

3. Creating a Jira Project

Created a Jira project for the Student Management System to organize and manage project activities.

Project Details:

- Project Name: Student Management System1
- Project Key: STUD
- Project Purpose: To manage student-related functionalities and testing activities.
- Testing Tool: Xray for Jira

Learned how to create a project, explore the project interface, and manage work items using Jira.

4. Understanding User Stories

4.1 What is a User Story?

A User Story describes a feature from the user's point of view.

4.2 Creating a User Story

Created a User Story for the Student Login functionality.

User Story Details:

- Issue Key: STUD-1
- Summary: Student Login Functionality
- Module: Student Login
- Priority: High

User Story:

As a registered student, I want to log in using my email and password so that I can access my student dashboard.

Acceptance Criteria:

- Student should be able to enter a registered email.
- Student should be able to enter the correct password.
- Login should be successful with valid credentials.
- Successful login should redirect to the student dashboard.

The User Story helps define the functionality that needs to be tested.

5. Understanding Preconditions

5.1 What is a Precondition?

A Precondition defines the conditions that must be satisfied before executing a test case.

5.2 Creating a Precondition

Created a Precondition for the Student Login test.

Precondition Details:

- Issue Key: STUD-2
- Summary: PRE01 - Student account must be registered

Description:

The student must have an existing registered account before attempting to log in.

Precondition Requirements:

- Student account must be registered.
- Student email must be valid.
- Student password must be available.

This Precondition was associated with the Student Login test case.

6. Understanding Xray Test Cases

6.1 What is a Test Case?

A Test Case contains a set of steps used to verify whether a particular functionality works correctly.

6.2 Creating an Xray Test

Created an Xray Test for verifying the Student Login functionality.

Test Details:

- Issue Key: STUD-3
- Summary: Verify student login with valid credentials
- Test Type: Manual
- Module: Student Login
- Priority: High

Test Description:

Verify whether a registered student can successfully log in using the correct email and password.

Test Steps:

Step	Action	Data	Expected Result
1	Open Student Management System login page	QA application URL	Login page should be displayed
2	Enter registered student email	student@gmail.com	Email should be accepted
3	Enter correct password	Student@123	Password should be accepted and masked
4	Click Login button	None	Student should successfully log in and dashboard should open

6.3 Linking Precondition with Test

Associated STUD-2 (Student account must be registered) with STUD-3.

This ensures that the required condition is satisfied before executing the test.

7. Understanding Test Sets

7.1 What is a Test Set?

A Test Set is a collection of related test cases grouped together for testing a particular functionality or module.

7.2 Creating a Test Set

Created a Test Set for organizing Student Login testing.

Test Set Details:

- Issue Key: STUD-4
- Summary: Student Login Test Suite
- Module: Student Login

Added the existing test STUD-3 to the Test Set.

The Test Set helps organize related test cases in one place.

8. Understanding Test Plans

8.1 What is a Test Plan?

A Test Plan is used to organize testing activities for a project or release.

It defines the testing scope and helps track the progress of testing activities.

8.2 Creating a Test Plan

Created a Test Plan for the Student Management System.

Test Plan Details:

- Issue Key: STUD-5
- Summary: Student Management System - Release 1.0 Testing
- Target Environment: QA
- Release: 1.0

Testing Scope:

- Student Login functionality.
- Verification of valid student credentials.
- Successful redirection to the student dashboard.

Added the existing test STUD-3 to the Test Plan.

The Test Plan helps organize and monitor the testing activities.

9. Understanding Test Execution

9.1 What is Test Execution?

Test Execution is the process of executing test cases and recording their actual results.

It helps determine whether the application behaves according to the expected results.

9.2 Creating a Test Execution

Created a Test Execution for Student Login testing.

Test Execution Details:

- Issue Key: STUD-6
- Summary: Student Login - QA Execution Cycle 1
- Assignee: Mouni Akshya Atkuri
- Test Environment: Not configured as a selectable environment.

Added the existing test STUD-3 to the Test Execution.

9.3 Executing the Test Case

Executed the Student Login test using the manual test execution interface.

The four test steps were marked as PASSED.

Execution Results:

Test Step	Expected Result	Status
Open login page	Login page displayed	PASSED
Enter registered email	Email accepted	PASSED
Enter correct password	Password accepted and masked	PASSED
Click Login button	Student dashboard opened	PASSED

Overall Test Execution Status: PASSED

The execution verified the Student Login functionality using the configured manual test steps.

10. Understanding Test Runs

10.1 What is a Test Run?

A Test Run represents an individual execution instance of a test case.

It contains execution details, test step results, and the overall test status.

10.2 Test Run Status

Common test execution statuses include:

- TODO: Test execution is not started.
- EXECUTING: Test is currently being executed.
- PASSED: Test execution is successful.
- FAILED: Test execution has failed.
- ABORTED: Test execution has been stopped.

The Student Login test run was completed with PASSED status.

11. Bug Reporting and Defect Management

11.1 What is a Bug?

A Bug is an error or defect in an application that causes it to behave differently from the expected result.

11.2 Bug Reporting

A Bug generally contains:

- Bug Summary
- Description
- Steps to Reproduce
- Expected Result
- Actual Result
- Priority
- Environment

11.3 Bug Workflow

A common bug lifecycle includes:

1. Open
2. In Progress
3. Fixed
4. Retest
5. Closed

In this project, the Student Login test passed successfully. Therefore, no Bug was created.

12. Understanding Xray Reports

12.1 What are Test Reports?

Test Reports provide information about test execution results, testing progress, and requirement coverage.

12.2 Report Generation

Explored the Jira Reports section and completed the report-generation activity for the project.

The reports help understand:

- Test execution status.
- Passed and failed tests.
- Test Plan progress.
- Requirement coverage.
- Testing activities.

Project Test Results:

Metric	Result
Total Test Cases	1
Passed	1
Failed	0
Pending	0
Overall Execution	PASSED

Reports help testing teams monitor the quality and progress of the application.

13. Hands-On Practice and Learning

During the practical session, I learned about:

1. Creating a Jira project.
2. Understanding the Jira project interface.
3. Creating a User Story.
4. Defining acceptance criteria.
5. Creating Preconditions.
6. Creating Xray Manual Tests.
7. Adding test steps and expected results.

8. Associating Preconditions with Tests.
9. Creating Test Sets.
10. Adding Tests to Test Sets.
11. Creating Test Plans.
12. Adding Tests to Test Plans.
13. Creating Test Executions.
14. Executing manual test cases.
15. Recording PASS and FAIL results.
16. Understanding Test Runs.
17. Understanding Bug Reporting and Defect Management.
18. Exploring Jira Reports and Xray reporting concepts.

14. Complete Xray Testing Workflow

Learned the overall testing workflow using Jira and Xray:

1. Create a Jira Project.
2. Create a User Story.
3. Define Acceptance Criteria.
4. Create a Precondition.
5. Create an Xray Test Case.
6. Associate the Precondition with the Test.
7. Create a Test Set.
8. Add the Test to the Test Set.
9. Create a Test Plan.
10. Add the Test to the Test Plan.
11. Create a Test Execution.
12. Add the Test to the Test Execution.
13. Execute the Test Steps.
14. Record Actual Results and Test Status.
15. Generate and review Test Reports.

Project Work Item Structure:

Issue Key	Work Item	Summary
STUD-1	Story	Student Login Functionality
STUD-2	Precondition	PRE01 - Student account must be registered
STUD-3	Test	Verify student login with valid credentials
STUD-4	Test Set	Student Login Test Suite
STUD-5	Test Plan	Student Management System - Release 1.0 Testing
STUD-6	Test Execution	Student Login - QA Execution Cycle 1

15. Conclusion

Through this practical training, I gained hands-on knowledge of Jira and Xray test management using the Student Management System project.

I learned how to create User Stories, define Preconditions, develop manual test cases, organize tests using Test Sets, prepare Test Plans, and execute test cases using Test Executions.

I also learned how to record test results, understand Test Runs, manage defects, and explore test reporting.

The practical implementation of the Student Login functionality helped me understand how Xray integrates with Jira to manage software testing activities.

The project successfully demonstrated the complete testing workflow using one manual test case, with all four test steps passing successfully.